THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. An element has atomic number 19. The number of valence electrons is: 1

(a) 8 **(b)** 2 **(c)** 1 **(d)** 9

2. The predictions of Newton's law of gravitation matched the observed motion of planets. This: 1

(a) supports the law **(b)** proves it forever
(c) refutes it **(d)** has no effect

3. Which technique separates a mixture of iodine and sand? 1

(a) magnet **(b)** sublimation **(c)** decantation **(d)** filtration

4. The formula of ammonium phosphate is: 1

(a) $(NH_4)_2PO_4$ **(b)** $(NH_4)_3PO_4$ **(c)** NH_4PO_4 **(d)** $NH_4(PO_4)_3$

5. An element with atomic number 14 has valency: 1

(a) 2 **(b)** 8 **(c)** 4 **(d)** 3

6. 72 km/h in m/s is: 1

(a) 20 **(b)** 7.2 **(c)** 200 **(d)** 72

7. In fractional distillation the component with the lowest boiling point: 1

(a) is collected first **(b)** stays in the flask
(c) is collected last **(d)** is not collected

8. Which of the following substances will conduct electricity in the solid state? 1

(a) sugar **(b)** sulphur **(c)** sodium chloride **(d)** copper

9. Boron has isotopes 10 (20 per cent) and 11 (80 per cent). The average atomic mass is: 1

(a) 10.2 u **(b)** 11 u **(c)** 10.5 u **(d)** 10.8 u

10. A car travels 90 km in 1.5 hours. Its speed in m/s is: 1

(a) 60 m/s **(b)** 6 m/s **(c)** 16.7 m/s **(d)** 90 m/s

11. The volume of ethanol in 250 mL of a 20 per cent (v/v) solution is: 1

(a) 50 mL **(b)** 200 mL **(c)** 20 mL **(d)** 5 mL

12. A compound has 40 per cent sulphur and 60 per cent oxygen by mass. 20 g of sulphur combines with how much oxygen? 1

(a) 30 g **(b)** 60 g **(c)** 12 g **(d)** 40 g

13. The atom with the configuration 2, 8, 8, 2 is: 1

(a) argon **(b)** calcium **(c)** potassium **(d)** magnesium

14. The 'ideal gas' model works best at: 1

(a) high pressure and low temperature **(b)** any condition

(c) low pressure and high temperature **(d)** none

15. To separate a mixture of kerosene and water we use: 1

(a) distillation **(b)** separating funnel

(c) evaporation **(d)** magnet

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is the thermometer bulb placed at the side arm opening in distillation? 2

17. Why is the formula of water not HO? Explain with valency. 2

18. Calculate the average atomic mass of an element with isotopes 63 u (70 per cent) and 65 u (30 per cent). 2

19. How is a hypothesis different from a prediction? 2

20. Explain why sea water boils at a higher temperature than pure water. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. An atom has 3 shells. The valence shell has 3 electrons. Find the atomic number and the valency of the element. 3

22. Explain how a graph helps to identify a mistake in a measurement. 3

23. Why is the temperature of the liquid constant during distillation of a pure liquid and rising during that of a mixture? 3

24. An element X forms an oxide X_2O_3 with 70 per cent X by mass. Find the atomic mass of X (O = 16). 3

OR

Explain how the Bohr model accounts for the atomic spectra, in brief. 3

25. The diameter of a hair is 0.07 mm. Express it in metres and in micrometres. 3

OR

How will you prepare 250 mL of a 10 per cent (v/v) solution of acetic acid? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain how scientists establish a theory (e.g. cell theory or atomic theory) through evidence over time with at least five steps. 5

OR

Explain, with reasons, how air is separated in industry, and the differences in the boiling points used. 5

27. A student heats 6.0 g of copper in air. The mass of the product copper oxide is 7.5 g. Find the mass of oxygen, the ratio of Cu : O and compare with the formula CuO (Cu = 63.5, O = 16). 5

OR

Describe the isotopes of carbon, their composition, use and the idea of carbon dating. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND